

User Stories

US-001 AI Agent CLI Query Processing

As a user, I want to submit questions through the CLI and receive AI-generated responses, so that my queries are correctly classified and processed by capabilities like calculator and datetime.

Description: The application should accept user questions through the CLI and return an AI-generated response. The agent should classify questions and route them to the appropriate processing capability, specifically incorporating calculator and datetime capabilities.

Acceptance Criteria:

- AC-001 The system shall accept a valid user question through the CLI and return a generated response. [Source: REQ-004, REQ-009]
- AC-002 The system shall classify supported queries and route them to the corresponding processing capability, including calculator and datetime handling. [Source: REQ-007, REQ-008, REL-003]

Business Rules:

- BR-001: Supported queries must be classified and routed to corresponding processing capabilities.

Dependencies:

- Calculator and datetime capability implementations.

Assumptions:

- Not specified in the provided input.

Edge Cases:

- Entering an empty or whitespace-only message is handled without unnecessary AI processing.
- Web search request failure results in an external search-service failure probable cause, recommending safe exception handling and a controlled no-results/error response.

US-002 AI Image Analyzer File Validation and Analysis

As a user, I want to upload supported image formats under 10 MB for AI-based image analysis, so that I receive object detection, scene analysis, image summaries, and OCR.

Description: The system should analyze images and identify objects, scenes, summaries, and text. It must support image formats and reject files larger than 10 MB before analysis.

Acceptance Criteria:

- AC-001 The requirement summary shall mention AI-based image analysis including object detection, scene analysis, image summary, and OCR. [Source: REQ-021, REQ-022, REQ-023, REQ-024, REQ-025]
- AC-002 The system shall support image formats and reject files larger than 10 MB before analysis. [Source: REQ-017, REQ-018, REQ-019]

Business Rules:

- BR-001: Files larger than 10 MB must be rejected before analysis.

Dependencies:

- AI image analysis service.

Assumptions:

- Not specified in the provided input.

Edge Cases:

- Uploading a PDF file instead of an image results in rejection with a controlled validation error.
- Handling an image larger than 10 MB that is incorrectly accepted.

US-003 Document Summary Assistant Upload and Processing

As a user, I want to upload PDF or DOCX documents up to 10 MB, so that the system extracts readable text and generates a summary.

Description: The system shall allow a user to upload a document for summarization, accepting PDF and DOCX documents while rejecting documents larger than the configured 10 MB limit, and extracting readable PDF text.

Acceptance Criteria:

- AC-001 User can submit a document for summarization, accepting PDF and DOCX uploads. [Source: REQ-033, REQ-034]
- AC-002 Documents exceeding 10 MB are rejected. [Source: REQ-035]
- AC-003 Readable PDF text is extracted for summarization. [Source: REQ-065]

Business Rules:

- BR-001: Accepted file formats are PDF and DOCX.
- BR-002: Maximum file size limit is 10 MB.

Dependencies:

- PDF text extraction service.

Assumptions:

- Not specified in the provided input.

Edge Cases:

- Submitting a document larger than 10 MB triggers rejection.

US-004 AI Email Generator Tone and Field Output

As a user, I want the AI generation service to return subject and body fields based on my bullet points and tone choices, so that I receive structured email drafts.

Description: The AI generation service shall return subject and body fields, supporting email generation from valid bullet points and using specified tones like Friendly.

Acceptance Criteria:

- AC-001 Successful generation contains subject and body fields. [Source: REQ-053]
- AC-002 Generated email uses the selected tone (e.g., Friendly) and is generated from valid bullet points. [Source: GE-247, GE-248]

Business Rules:

- BR-001: The AI generation service response must include subject and body fields.

Dependencies:

- AI generation service.

Assumptions:

- Not specified in the provided input.

Edge Cases:

- Generation with missing or invalid bullet points.

US-005 Smart FAQ Bot PDF Upload, Extraction, and Chunking

As a user, I want to upload PDF files up to 25 MB, so that the system extracts readable text and page metadata using PyMuPDF and splits the text into chunks.

Description: The system shall accept only PDF files, reject non-PDF uploads with a controlled error, reject PDF files larger than 25 MB, extract text from PDFs using PyMuPDF yielding readable page text and page metadata, and split extracted PDF text into chunks using configured size and overlap.

Acceptance Criteria:

- AC-001 Non-PDF uploads are rejected with a controlled error. [Source: REQ-066]
- AC-002 Files above 25 MB are rejected. [Source: REQ-067]
- AC-003 Readable page text and page metadata are extracted using PyMuPDF. [Source: REQ-068]
- AC-004 Chunks are generated using configured size and overlap. [Source: REQ-069]

Business Rules:

- BR-001: Only PDF files are accepted.
- BR-002: Maximum PDF file size is 25 MB.

Dependencies:

- PyMuPDF library.

Assumptions:

- Not specified in the provided input.

Edge Cases:

- Rejecting a PDF larger than 25 MB returns a controlled file-size error.

US-006 Meeting Transcript & Minutes Generator System and Ingestion

As a user, I want the system to expose a health endpoint and process transcripts or media recordings, so that I can generate structured meeting minutes from valid inputs.

Description: The system shall expose a health endpoint (GET / returns status healthy with HTTP 200). It shall accept a meeting transcript as text, allow TXT transcript uploads, reject non-TXT transcript uploads with controlled errors or HTTP 400 invalid-format responses, and accept supported audio and video recordings (such as MP4).

Acceptance Criteria:

- AC-001 GET / returns status healthy with HTTP 200. [Source: REQ-081]
- AC-002 A non-empty transcript can be submitted for minutes generation, and valid TXT content is returned as transcript text. [Source: REQ-082, REQ-083]
- AC-003 Invalid transcript file types return controlled errors or HTTP 400 invalid-format response. [Source: REQ-084, TC-306]
- AC-004 Supported media extensions (such as MP4) are accepted and can enter the transcription flow. [Source: REQ-085, TC-308]

Business Rules:

- BR-001: Only supported TXT transcripts and supported audio/video extensions are accepted.

Dependencies:

- Transcription and minutes generation service.

Assumptions:

- Not specified in the provided input.

Edge Cases:

- Rejecting a PDF as a transcript upload returns an HTTP 400 invalid-format response.